

Harshitha Puttaswamy

Binghamton, NY (Willing to relocate) | (716) 612-0339 | harshitha2680@gmail.com
linkedin.com/in/harshitha-p | github.com/harshithaputtaswamy | harshithaputtaswamy.github.io/portfolio/

PROFESSIONAL SUMMARY

- Skilled Software Engineer with experience in full-stack development, backend engineering, and AI-driven applications, with a strong foundation in algorithms, reinforcement learning, and cloud infrastructure.
- Proficient in Python, TypeScript, and C, and experienced in building scalable systems using Django, MERN stack, NestJs, NextJs, PostgreSQL, and REST APIs, adept in CI/CD pipelines (GitHub Actions, Docker, Kubernetes), and capable of bridging research and production systems.
- Demonstrated expertise in applied AI research, including Multi-Armed Bandits, Deep Q-Learning, and LLM-based decision modeling, implemented a parallelized, multi-GPU inference framework using Ollama for large-scale fairness analysis, significantly accelerating research workflows.

PROFESSIONAL EXPERIENCE

Research Assistant, Analysis of Envy-Freeness of Allocation Algorithms Using LLMs

February 2024 - Present

Binghamton University, Department of Computer Science, Binghamton, NY, USA

- Developed and implemented a scalable framework to analyze the envy-freeness of item allocations for n agents using different algorithms, evaluating responses with large language models (LLMs)
- Designed prompt engineering techniques to elicit agent preferences and measure swap-based reasoning; applied chi-square tests to assess statistical independence across algorithms and agent personality traits, with validation against MTurk data.
- Implemented a multi-threaded inference system using multiple Ollama instances, distributing workloads across GPUs to evaluate multiple allocation strategies concurrently, significantly improving evaluation speed and enabling real-time experimental scaling.

Research Assistant, Iterative Voting for Committee Selection with Multi-Armed Bandits

January 2024 - December 2024

Binghamton University, Department of Computer Science, Binghamton, NY, USA

- Developed and implemented a dynamic Multi-Armed Bandit (MAB) framework for iterative decision-making, optimizing voter utilities, and drawing parallels to complex bidding strategies in ad platforms as part of the Iterative Voting for Committee Selection with Multi-Armed Bandits project
- Engineered scalable algorithms for strategic optimization in multi-agent systems, leveraging Integer Linear Programming (ILP) to solve resource allocation and ranking problems akin to budget distribution
- Conducted extensive experimental analysis on preference aggregation and rule convergence, producing insights on maximizing utility and satisfaction, applicable to auction mechanisms and real-time bidding environments

Fyle Technologies Ltd, Backend Developer, Internship | Bangalore, India

March 2022 - December 2022

- Developed and optimized APIs using Django, Angular, and PostgreSQL (PSQL), contributing to 8 new features, improving the product's market appeal
- Enhanced code quality by introducing unit and integration tests using Pytest (98%+ coverage) across multiple projects
- Introduced GitHub Actions for Continuous Integration, enabling automatic execution of tests for each pull request, reducing review time by 90%

PROJECT EXPERIENCE

Splitwiser – A Splitwise clone | Full Stack Development, Amazon Bedrock

June 2024 - December 2024

- Created a Splitwise-inspired application featuring receipt scanning via camera or file upload, group and non-group expense tracking, and dynamic graphical and manual split functionalities
- Leveraged Amazon Bedrock API to convert receipt images into JSON data, to facilitate graphical expense visualization
- Utilized NextJS for UI and API development, Google Firebase for backend services, and Firestore for storing receipt data

Autokube | DevOps, CI/CD

August 2021 - March 2022

- Led the development of the web application for automating Kubernetes cluster creation and modification
- Utilized MongoDB, Express.js, React, Node.js (MERN) stack for formulating APIs to initiate scripts, triggering a series of workflows for managing the backend Kubernetes clusters

EDUCATION

Binghamton University, State University of New York

Master of Science in Computer Science

December 2024

TECHNICAL SKILLS

Programming Languages and DB: C, Python, Java, JavaScript, TypeScript, MongoDB, PostgreSQL, MYSQL, Firebase

Frameworks and Software's: AWS, Amazon Bedrock, AWS Lambda, Dynamo DB, EC2, ExpressJS, ReactJS, NodeJS, NestJS, NextJS, Django, CI/CD, Pytest, Git, Linux, Kubernetes, TensorFlow, Keras, Pytorch, Pandas, Google Colab